

TENBEL OFFGRID TECHNICAL OVERVIEW

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OVERVIEW OF TENBEL OFFGRID



Sub-GHz LoRa Technology

Utilizes long-range, low-power sub-GHz LoRa radio for reliable communication without traditional networks.

Smartphone-Enabled Messaging

Messaging through a dedicated mobile app with smartphone as user interface connected via BLE or USB.

Resilient Infrastructure-Free Use

Designed for disaster, rural, and remote areas where cellular, satellite, or Wi-Fi networks are unavailable.

Flexible Networking Architecture

Supports peer-to-peer, gateway-assisted, and multi-hop mesh networking for extended coverage and reliability.

DESIGN PRINCIPLES



Infrastructure Independence

OffGrid operates without cellular, Wi-Fi, or satellite connectivity, enabling use during disasters and in remote areas.

Smartphone-First User Experience

Users interact via a familiar smartphone app, simplifying messaging while RF communication runs on LoRa hardware.

Long-Range RF Reliability

Sub-GHz LoRa CSS modulation ensures robust, low-power, long-distance communication in noisy environments.

Flexible Network Deployment

Supports peer-to-peer, gateway-extended, or multi-hop mesh networks for versatile coverage in tough terrains.

SYSTEM COMPONENTS AND ARCHITECTURE

SYSTEM COMPONENTS



Smartphone User Interface

The smartphone runs an offline mobile app with a chat-style interface for composing and receiving messages.



Messaging Node Hardware

The Messaging Node uses LoRa radio, connects via Bluetooth or USB, and manages radio communication efficiently.



LoRa Gateway/Relay Node

Optional gateways extend coverage and support communication over larger or obstructed areas, either fixed or portable.

HIGH-LEVEL ARCHITECTURE



User Interaction Isolation

The architecture separates user interaction from RF communication for simplicity and durability in messaging.

Long-Range LoRa Communication

Sub-GHz LoRa radio enables low-power, long-range communication between nodes, supporting diverse deployment models.

Self-Contained Network Design

The system operates without backbone infrastructure, servers, or cloud services, ensuring independence and reliability.

Simplified User Experience

Abstracting RF details reduces smartphone battery use and eases training for field operators.

MESSAGING FLOW AND RF PRINCIPLES

MESSAGING FLOW



User Message Composition

Users compose messages on the OffGrid smartphone app, creating a simple and low-power interface experience.

Message Transfer to LoRa Node

Messages are transferred via Bluetooth Low Energy or USB to the energy-efficient LoRa Messaging Node for transmission.

LoRa RF Transmission

The Messaging Node broadcasts the message as a LoRa RF signal over sub-GHz spectrum, enabling long-distance communication.

Relay and Reception Process

Relay nodes forward messages beyond direct range, ensuring reliable delivery even in infrastructure outages.

RF FUNDAMENTALS



LoRa CSS Modulation

LoRa CSS modulation enables communication at very low signal-to-noise ratios using wideband chirp signals.

Sub-GHz Frequency Benefits

Operating in sub-GHz bands allows signals to travel longer distances and better penetrate obstacles.

Efficient Low-Power Design

High link budget efficiency supports battery-powered nodes requiring long unattended operation.

Reliability over Throughput

Focus on reliable, short message transmission suits emergency and rural communication needs.

RF DEPLOYMENT MODELS AND PERFORMANCE

RF DEPLOYMENT MODELS



Open Terrain Deployment

In open or rural areas, long line-of-sight communication allows ranges of 10 to 15 kilometers using appropriate antennas.

Mountainous Deployment

Relay nodes placed along ridgelines help overcome terrain shadows ensuring reliable coverage in mountainous environments.

Urban and Disaster Deployment

Portable relay nodes and planned gateways help bypass obstructions in dense urban and disaster areas to maintain connection.

Multi-Hop Mesh Networking

Multi-hop mesh networks enable messages to traverse multiple nodes, useful in forests, mining corridors and border areas.

PERFORMANCE CONSIDERATIONS



Optimized for Lightweight Messaging

OffGrid uses LoRa technology focusing on energy efficiency and range over high data throughput.

Latency and Reliability

Latency ranges in seconds, suitable for coordination and safety-critical alerts under tough RF conditions.

Scalability by Node Addition

Coverage extends by adding nodes or relays, avoiding bandwidth increase and maintaining efficiency.

Power Efficiency and Durability

Design balances power conservation and dependable operation in challenging terrains over long durations.

CONCLUSION

SUMMARY OF TENBEL OFFGRID



Infrastructure-Independent Communication

Tenbel OffGrid enables reliable messaging without cellular, satellite, or Wi-Fi networks using dedicated LoRa nodes and smartphone apps.

Flexible Network Architecture

Supports peer-to-peer, gateway-assisted, and multi-hop mesh networking for adaptable deployment in varied environments.

Low Power and Durable Design

Utilizes Chirp Spread Spectrum modulation for long range, RF resilience, and low power consumption suitable for field devices.

Ideal for Challenging Environments

Perfect for disaster response, remote exploration, rural areas, and industrial operations requiring dependable communication.